

Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

AMOT

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- Rising ransomware threats in cyber security
- Traditional antivirus limit actions
- Need for intelligent malware detection
- Multi-layered analytical approach



Ransomware attacks are becoming increasingly advanced and difficult to detect using traditional antivirus systems. This project provides an intelligent malware analysis solution using machine learning.

- Signature-based systems fail against zero-day malware
- Large number of false-positive alerts
- Manual malware analysis is time-consuming
- Need for automated ransomware analysis

Traditional antivirus systems mainly rely on signature-based detection methods, which are ineffective against modern ransomware, zero-day attacks, and polymorphic malware. There is a need for an intelligent and automated system that can accurately detect, analyze, and classify malicious files using multiple advanced detection techniques.



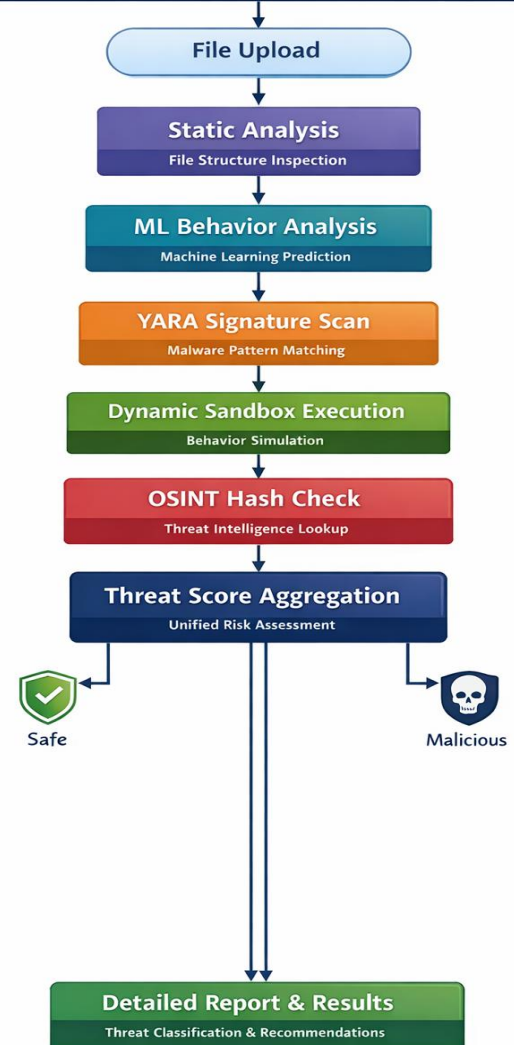
- Detect ransomware using multiple techniques
- Integrate ML, YARA, and OSINT
- Reduce malware analysis time
- Generate forensic reports automatically



AMOT: Ransomware & Forensics Analyser is an intelligent cybersecurity platform designed to detect and analyse ransomware threats using machine learning, YARA scanning, and OSINT verification. The system automates malware analysis and generates detailed forensic reports for faster and more accurate threat detection.

- User uploads a suspicious file through the dashboard.
- The system performs static analysis and extracts file features.
- Machine learning predicts malicious behavior of the file.
- YARA rules scan for known ransomware signatures.
- OSINT APIs verify file reputation using threat intelligence databases.
- Final threat score and forensic report are generated and displayed.

AMOT Ransomware & Forensics Analyser



Frontend Layer

Developed using React.js.

Provides user dashboard and history management.

Backend Layer

Developed using Python and Fast API.

Performs malware analysis and API handling.

Integrates machine learning and YARA engines.

Database Layer

SQLite database stores reports and metadata.

Maintains analysis history and logs.

Technology Used

Python 3.10	Backend development
Fast API	API framework
React.js	Frontend development
SQLite	Database management
YARA-Python	Malware signature scanning
Pandas	Data processing
NumPy	Numerical analysis
Requests	OSINT API integration
Framer Motion	Frontend animations



AMOT AUTHORIZED

The AMOT Triage Node is now synchronized with your environment.
All forensic modules are standing by for payload detonation.

STATIC FORENSICS

READY

ML CLASSIFIER

ACTIVE

OSINT SYNC

ONLINE

LAUNCH AMOT LAB →



 **AMOT**
SECURITY LAB

 **Cyber Triage**

 Intel Briefing

 **GLOBAL PROTECTED**
Version: 1.0.4 Enterprise
Node: PRD-NYC-01

Cyber Threat Triage

Multi-dimensional hybrid analysis of suspicious binaries.

ENGINE STATUS: ONLINE



Ingest Payload

Drop any suspicious PE or binary for global intelligence lookup and deep static forensics.

SELECT FILE
SYSTEM



More
Info

Session Intel



EMPTY - PLEASE UPLOAD FILE FIRST

TTP Matrix

MITRE ATT&CK Technical mapping.

NO UPLOAD DETECTED

Threat Report

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**LIKELY
BENIGN**

4%

GLOBAL THREAT SCORE: 4%



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Cyber Triage



Intel Briefing

CONSENSUS: High (Baseline Verified)

TITANIUM-ENSEMBLE (RF + GBM + ET) | 3 MODELS
SYNCED

SIG MATCHES

0 Ransomware Rules



FILE IS SAFE

Risk assessment is low. No malicious behaviors or ransomware signatures were detected.

SAFETY VERIFICATION

Our multi-stage analysis (Static, ML, and OSINT) confirms this file does not match known ransomware families. It has a valid structure and does not attempt to illegally encrypt files or hijack system processes.



Global Intelligence & History

LIVE OSINT SYNC

AV DETECTIONS

0 / 0

Source: VirusTotal

THREAT FAMILY

Undefined

ML Cluster Mapping

LAST SCAN DATE

**Recently
Identified**

Global Sync Time

localhost:5175 says

AMOT Guard 1.0.4 Enterprise:

Developed for advanced ransomware identification.

Upload any suspicious binary to trigger 5-stage hybrid analysis.

OK



- AI-powered ransomware and malware detection
- YARA-based signature and pattern scanning
- Real-time OSINT threat intelligence verification
- Automated forensic report generation
- Fast, scalable, and user-friendly analysis platform

Future Scope



- Integration of deep learning-based malware detection
- Advanced sandbox behavioral analysis
- Real-time network traffic monitoring
- Cloud deployment for scalability
- Automated incident response system
- Enterprise-level database support
- Improved machine learning accuracy with larger datasets



prediction, YARA rule matching, and OSINT-based

threat intelligence, the system improved detection accuracy and reduced malware investigation time.

The project also generated detailed forensic reports and provided a user-friendly solution for cybersecurity analysts, making it a scalable and effective platform for modern cybersecurity environments.

Conclusion

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Thank You